Spaceship in the atmosphere — Solution

X25 (2025) — English translation

A1 **0.20 pts**

First, we obtain the exact equations that describe the dynamics of the spacecraft. Express the time derivatives of altitude h and range s in terms of v, γ, r_0, r.

The rate of decrease in height is the vertical projection of speed $v\sin\gamma$, the speed of horizontal movement $v\cos\gamma$ after projection onto the surface of the Earth decreases by r/r_0 times.

Answer:

$$\dot{h} = -v\sin\gamma, \quad \dot{s} = v\cos\gamma \cdot r_0/r$$

A2 **1.00 pts**

Find the derivatives of the velocity modulus v and angle γ. The answer may also include the lift force L, the air resistance force D, the acceleration due to gravity at a given height g, the mass of the spacecraft m, v, γ, r.

Newton's second law in projection onto the axes along and perpendicular to the speed:

$$ma_\tau = -D + mg\sin\gamma, \quad ma_n = -L + mg\cos\gamma.$$

Tangential acceleration $a_\tau = \dot{v}$, therefore

$$a_\tau = -\frac{D}{m} + g\sin\gamma.$$

Normal acceleration $a_n = \omega v$, where ω – angular velocity rotation vector velocity, which consider/count positive if the angular velocity rotates downward. This velocity not equal derivative angle γ, since direction horizontal change from point to point with angular velocity $\omega_h = v\cos\gamma/r$, then

$$\omega = \dot{\gamma} + \frac{v\cos\gamma}{r},$$

therefore equation for normal acceleration

$$mv\left(\dot{\gamma} + \frac{v\cos\gamma}{r}\right) = -L + mg\cos\gamma,$$

where do we find the required derivative

Answer:

$$\dot{v} = -\frac{D}{m} + g\sin\gamma, \quad \dot{\gamma} = -\frac{L}{mv} + \frac{g}{v}\cos\gamma - \frac{v\cos\gamma}{r}$$

B1 **0.40 pts**

Obtain an expression for the derivative dv/dh. Represent the answer as

$$\frac{dv}{dh} = Af(v,\, h),$$

where A – constant, which can depend from all constant, characterizing space spacecraft and atmosphere (m, S, C_D, C_L, ρ_0, β, γ), and f – some function height and velocity. in further use constant **A** *to write answers.*

Let us substitute the expression for the air resistance force D into the equation of motion and obtain, neglecting gravitational acceleration

$$\dot{v} = -\frac{\rho_0 S C_D}{2m} v^2 e^{-\beta h}.$$

Taking into account formula for $\dot{h} = -v \sin\gamma$ obtain

$$\frac{dv}{dh} = \frac{\dot{v}}{\dot{h}} = \frac{\rho_0 C_D S}{2m \sin\gamma} \cdot v e^{-\beta h}$$

Answer:

$$\frac{dv}{dh} = A v e^{-\beta h}, \quad A = \frac{\rho_0 C_D S}{2m \sin\gamma}$$

B2 0.80 pts

Find the dependence of the speed of the spacecraft on altitude. At the initial height h_0 the speed was equal to v_0. Express your answer in terms of A, β, h, v_0, h_0. Also get an approximate answer, assuming that at an altitude of h_0 the atmospheric density is negligible (in this case the answer should not include h_0). In the following, use an approximate expression throughout.

Dividing the variables, we get

$$\frac{dv}{v} = A e^{-\beta h} dh,$$

integrate, obtain

$$\int_{v_0}^{v} \frac{dv}{v} = \int_{h_0}^{h} A e^{-\beta h} dh, \quad \ln\frac{v}{v_0} = -\frac{A}{\beta}(e^{-\beta h} - e^{-\beta h_0}).$$

Hence find expression for velocity

$$v = v_0 \exp\left(-\frac{A}{\beta}(e^{-\beta h} - e^{-\beta h_0})\right)$$

For sufficiently larger value h_0 term $e^{-\beta h_0} \ll e^{-\beta h}$ one can neglect

$$v = v_0 \exp\left(-\frac{A}{\beta} e^{-\beta h}\right)$$

Answer:

$$v = v_0 \exp\left(-\frac{A}{\beta}(e^{-\beta h} - e^{-\beta h_0})\right), \quad v \approx v_0 \exp\left(-\frac{A}{\beta} e^{-\beta h}\right)$$

B3 0.30 pts

Find the dependence of the acceleration a (that is, the derivative of the velocity modulus) of the spacecraft on altitude. Express your answer in terms of v_0, A, β, h, γ.

Let us substitute the formula for velocity into the equation for tangential acceleration and obtain

$$a = \dot{v} = -\frac{D}{m} = -\frac{\rho_0 S C_D}{2m} v^2 e^{-\beta h} = -A v_0^2 \sin\gamma \exp\left(-\frac{2A}{\beta} e^{-\beta h} - \beta h\right)$$

Answer:

$$a = -A v_0^2 \sin\gamma \exp\left(-\frac{2A}{\beta} e^{-\beta h} - \beta h\right)$$

B4 **0.50 pts**

Find the height h_c at which the acceleration modulus is maximum. Obtain a formula for the maximum value of acceleration modulus a_{max} and velocity v_c at height h_c. Express your answers using A, β, γ, v_0.

The acceleration module is maximum, with a minimum exponent,

$$\frac{d}{dh}\left(-\frac{2A}{\beta}e^{-\beta h}-\beta h\right)=2Ae^{-\beta h}-\beta=0.$$

Hence critical value height

$$h_c=\frac{1}{\beta}\ln\frac{2A}{\beta}.$$

Substitute this value height in formula for height and acceleration, obtain

$$a_{max}=Av_0^2\sin\gamma\exp\left(-\frac{2A}{\beta}e^{-\beta h_c}-\beta h_c\right)=\frac{\beta v_0^2\sin\gamma}{2e}$$

$$v_c=v_0\exp\left(-\frac{A}{\beta}e^{-\beta h_c}\right)=v_0e^{-1/2}$$

Answer:

$$h_c=\frac{1}{\beta}\ln\frac{2A}{\beta},\quad v_c=v_0e^{-1/2},\quad a_{max}=\frac{\beta v_0^2\sin\gamma}{2e}$$

B5 **0.60 pts**

For the given numerical data, find the speed (in km/s) and acceleration (in units of g_0) at heights $h_1=80$ km, $h_2=60$ km, $h_3=40$ km.

Substituting the height values into the formulas for speed and acceleration, we get

Answer: h, km a/g_0 v, km/s 80 -0.20 7.7 60 -2.8 7.2 40 -5.9 2.7

B6 **0.20 pts**

Find the maximum acceleration (in units of g_0) and the critical height h_c at which it is achieved.

Answer:

$$a_{max}=8.6\ g_0,\quad h_c=45.5\text{ km}$$

C1 **0.10 pts**

Find the first escape velocity v_s – the speed of the spacecraft in a circular orbit, the radius of which is equal to the radius of the Earth. Express your answer in terms of r_0, g_0.

Let us write the equation of motion for acceleration when moving in a circle

$$\frac{v^2}{r_0}=g_0,$$

Answer:

$$v_s=\sqrt{g_0r_0}$$

C2 **0.40 pts**

Determine the speed at which the spacecraft must move at altitude h for the motion described in the introduction to this part. Express your answer in terms of v_s, ρ_0, β, C_L, S, m, r_0, h.

Let us write the equation for normal acceleration, taking into account that the derivative $\dot{\gamma}$ is small, and $\cos\gamma \approx 1$ and it can be neglected

$$-\frac{L}{m} + g_0 - \frac{v^2}{r_0} = 0.$$

Substitute formula for lift force, and also for dependence density air from height, obtain

$$v^2\left(\frac{1}{r_0} + \frac{C_L S \rho_0}{2m} e^{-\beta h}\right) = g_0,$$

from which find

$$v^2 = \frac{g_0 r_0}{1 + \dfrac{C_L \rho_0 r_0 S}{2m} \cdot e^{-\beta h}}$$

Answer:

$$v = \frac{v_s}{\sqrt{1 + \dfrac{C_L \rho_0 r_0 S}{2m} \cdot e^{-\beta h}}}$$

C3 — 0.40 pts

Determine the acceleration a (derivative of the velocity modulus) created by the force of air resistance during such movement. Express your answer in terms of v, C_L, C_D, r_0, g_0.

Lift and air resistance are related by the relation

$$\frac{D}{L} = \frac{C_L}{C_D}$$

$$\dot{v} = -D/m = -\frac{C_D}{C_L m} \cdot L = -\frac{C_D}{C_L}\left(g_0 - \frac{v^2}{r_0}\right)$$

Answer:

$$a = -\frac{C_D}{C_L}\left(g_0 - \frac{v^2}{r_0}\right)$$

C4 — 0.50 pts

Let the initial speed of the spacecraft be $v_1 < v_s$ and the final speed $v_2 < v_1$. Find the horizontal displacement of the ship s during the motion. Consider that the entire change in the velocity module occurs due to the force of air resistance. Express your answer using r_0, C_L, C_D, v_s, v_1, v_2.

Using the relations $a = dv/dt$ and $v = ds/dt$ (in the approximation $\cos\gamma \approx 1$), we obtain the connection between displacement and change in speed,

$$\frac{ds}{dv} = \frac{v}{a} = -\frac{C_L}{C_D}\frac{v}{g_0 - v^2/r_0} = -\frac{C_L}{C_D} r_0 \frac{v}{v_s^2 - v^2}.$$

Integrate, obtain

$$s = -\frac{C_L}{C_D} r_0 \int_{v_1}^{v_2} \frac{v dv}{v_s^2 - v^2} = -\frac{C_L}{2C_D} r_0 \int_{v_1}^{v_2} dv\left(\frac{1}{v_s - v} + \frac{1}{v_s + v}\right) = \frac{C_L}{2C_D} r_0 \ln\frac{v_s^2 - v_2^2}{v_s^2 - v_1^2}.$$

Answer:

$$s = \frac{C_L}{2C_D} r_0 \ln\frac{v_s^2 - v_2^2}{v_s^2 - v_1^2}$$

C5 0.60 pts

Let the following parameters be given for the spaceship: $m = 84 \times 10^3$ *kg* $S = 250$ m^2 $C_D = 0.8$ $C_L = 0.9$ *Find the values of speed (in km/s) and acceleration due to air resistance (in units of* g_0*) at altitudes* $h_1 = 80$ *km,* $h_1 = 60$ *km,* $h_1 = 40$ *km.*

Find the values of speed (in km/s) and acceleration due to air resistance (in units of g_0) at heights $h_1 = 80$ km, $h_1 = 60$ km, $h_1 = 40$ km.

Using formulas for speed and acceleration expressed in terms of speed, we obtain

Answer: $h,$ km a/g_0 $v,$ km$/c$ 80 -0.13 7.3 60 -0.65 4.1 40 -0.87 1.2

C6 0.20 pts

Under the conditions of the previous paragraph, calculate the range of horizontal movement of the spacecraft when lowering from height $h_0 = 90$ *km to height* $h_f = 30$ *km.*

Let's calculate the initial and final speeds of the spacecraft

$$v_1 = v(h_0) = 7.7 \text{ km/s}, \quad v_2 = v(h_f) = 0.61 \text{ km}/c,$$

then the range of movement

Answer:

$$s = 11.4 \cdot 10^3 \text{ km}$$

D1 0.40 pts

Write down expressions for energy E and angular momentum L. Express your answer using G, m, a, e, M*. Note: It may be convenient to write the ZSE at the apocentric or periapsis point.*

Note: It may be convenient to write the ZSE at the apocentric or periapsis point.

Let the distances from the ground to the ship at the pericenter (the closest point) and apocenter (the most distant point) is equal to to r_1 and r_2, respectively. The speeds of the ship at these points are $L/(mr_1)$ and $L/(mr_2)$, respectively. Then:

$$E = \frac{L^2}{2mr_1^2} - \frac{GMm}{r_1} = \frac{L^2}{2mr_2^2} - \frac{GMm}{r_2},$$

Express hence angular momentum L:

$$L^2 = 2mr_1^2 E_1 + 2GMm^2 r_1 = 2mr_2^2 E_2 + 2GMm^2 r_2.$$

Hence express energy

$$E(r_2^2 - r_1^2) = GMm^2(r_1 - r_2),$$

and means

$$E = -\frac{GMm}{r_1 + r_2} = -\frac{GMm}{2a}.$$

Such same manner express angular momentum

$$\frac{L^2}{2m}\left(\frac{1}{r_1^2} - \frac{1}{r_2^2}\right) = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right), \quad L^2 = 2GMm^2 \frac{r_1 r_2}{r_1 + r_2}.$$

Express distance through larger semi-axis and eccentricity $r_1 = a(1-e)$, $r_2 = a(1+e)$ find

$$L^2 = GMm^2 a(1 - e^2).$$

Answer:

$$E = -\frac{GMm}{2a}, \quad L = \sqrt{GMm^2 a(1 - e^2)}$$

D2 — 0.20 pts

Write down the moment equation and expression for the power of the friction force P. Express your answer using $\vec{r}, \vec{v}, \alpha$.

Answer:

$$\frac{d}{dt}\vec{L} = -\alpha[\vec{r}, \vec{v}]$$

$$P = -\alpha v^2$$

D3 — 0.20 pts

Obtain an expression for the dependence of $\vec{L}$ on t if at time $t = 0$ $\vec{L} = \vec{L}_0$. Express your answer using α, m, $\vec{L}_0$.

Let's use the definition for $\vec{L}$ and write the moment equation in the form

$$\frac{d}{dt}\vec{L} = -\frac{\alpha}{m}\vec{L},$$

Answer:

$$\vec{L} = \vec{L}_0 \exp\left(-\frac{\alpha}{m}t\right)$$

D4 — 0.30 pts

Show that:

$$[\vec{a}, \vec{L}] = \beta \frac{d}{dt}\vec{e}_i$$

Where $\vec{a}$ – acceleration rocket, $\vec{e}_i$ – some unit vector polar coordinate with beginning/origin in center Earth. Find $\vec{e}_i$ and β. Note: Can be convenient use expression $\vec{L} = mr^2\vec{\omega}$.

Note: It may be convenient to use the expression $\vec{L} = mr^2\vec{\omega}$.
Newton's second law is

$$m\vec{a} = -\frac{GmM}{r^2}\vec{e}_r.$$

Substitute acceleration in vector product:

$$[\vec{a},\ \vec{L}] = -\left[\frac{GM}{r^2}\vec{e}_r,\ \vec{L}\right] = -GM\left[\frac{1}{r^2}\vec{e}_r,\ mr^2\vec{\omega}\right] = GmM\,[\vec{\omega},\ \vec{e}_r] = GmM\frac{d}{dt}\vec{e}_r.$$

Answer:

$$\beta = GMm, \quad \vec{e}_i = \vec{e}_r$$

D5 — 0.30 pts

Multiplying expression (1) by $\vec{r}$ scalarly and using the results of point D4, obtain the dependence of the modulus of the radius vector $\vec{r}$ on φ – the angle between vectors $\vec{A}$ and $\vec{e}_r$. Express your answer using L, m, G, e, φ, M.

Multiply the expression by $\vec{r}$, we get

$$(\vec{r}, [\vec{v}, \vec{L}]) = GMmr + \vec{A}\cdot\vec{r}.$$

in left part rearrange factor, use property mixed product:

$$(\vec{r}, [\vec{v}, \vec{L}]) = (\vec{L}, [\vec{r}, \vec{v}]) = \frac{L^2}{m}.$$

Then obtain

$$\frac{L^2}{m} = GmMr(1 + e\cos\varphi),$$

where we find

Answer:

$$r = \frac{L^2/GMm^2}{1 + e\cos\varphi}$$

D6 — 0.50 pts

Using expression (1) and the results of point D4, obtain the dependence of the square of the speed v^2 on φ. Express your answer using L, G, m, e, φ, M.

Let's square expression (1):

$$[\vec{v},\ \vec{L}]^2 = v^2L^2 = (GmM)^2 + 2GmM\vec{e}_r\cdot\vec{A} + A^2 = (GmM)^2(1 + 2e\cos\varphi + e^2),$$

Answer:

$$v^2 = \frac{(GMm)^2}{L^2}(1 + e^2 + 2e\cos\varphi)$$

D7 — 0.40 pts

Obtain an expression for the time-average squared velocity $\langle v^2\rangle$ as an integral over the angle φ. Express your answer in terms of $L, m, e, M, G, a, \varphi, \tau$, the rocket's orbital period. Note: It may be convenient to express the square of the average velocity from the averaged ZSE ($\langle T + U\rangle = \langle E\rangle$, where T is kinetic energy, U is potential).

Note: It may be convenient to express the square of the average velocity from the averaged ZSE ($\langle T + U\rangle = \langle E\rangle$, where T is kinetic energy, U is potential energy).

Method 1: By definition, the mean square of velocity is expressed through the integral as (τ – period of motion)

$$\langle v^2\rangle = \frac{1}{\tau}\int_0^\tau v^2 dt.$$

Differential time and angle φ one can relate with means moment momentum:

$$L = mr^2\frac{d\varphi}{dt}.$$

Transition to integration by angle

$$\langle v^2\rangle = \frac{1}{\tau}\int_0^\tau v^2\frac{dt}{d\varphi}d\varphi = \frac{m}{\tau L}\int_0^{2\pi} v^2r^2d\varphi.$$

Substitute expression from D5, D6, obtain

$$\langle v^2\rangle = \frac{L}{m\tau}\int_0^{2\pi}\frac{1 + e^2 + 2e\cos\varphi}{(1 + e\cos\varphi)^2}d\varphi.$$

. Method 2: From the time-averaged energy conservation we obtain:

$$\langle T\rangle = \frac{m}{2}\langle v^2\rangle = \langle E\rangle - \langle U\rangle = E + \left\langle\frac{GMm}{r}\right\rangle$$

Write expression for mean by time $1/r$ in form integral by angle:

$$\left\langle\frac{1}{r}\right\rangle = \frac{1}{\tau}\int_0^\tau \frac{1}{r}dt = \frac{1}{\tau}\int_0^\tau \frac{1}{r}dt\frac{dt}{d\varphi}d\varphi = \frac{m}{\tau L}\int_0^{2\pi} rd\varphi$$

Substitute still in expression for $\langle v^2\rangle$:

$$\langle v^2\rangle = \frac{2}{m}\left(E + \frac{L}{\tau}\int\limits_0^{2\pi}\frac{d\varphi}{1+e\cos\varphi}\right)$$

Answer:

$$\langle v^2\rangle = \frac{L}{m\tau}\int\limits_0^{2\pi}\frac{1+e^2+2e\cos\varphi}{(1+e\cos\varphi)^2}d\varphi$$

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D8 1.00 pts

Obtain an expression for $\langle\dot{E}\rangle$. Express your answer using G, m, a, e, M.

Method 1:

$$\langle v^2\rangle = \frac{L}{m\tau}\int\limits_0^{2\pi}\frac{1+e^2+2e\cos\varphi}{(1+e\cos\varphi)^2}d\varphi$$

. Let's use the universal trigonometric substitution:

$$\int\limits_0^{2\pi}\frac{1+e^2+2e\cos\varphi}{(1+e\cos\varphi)^2}d\varphi = 2\int\limits_0^{+\infty}\frac{1+e^2+2e(1-t^2)/(1+t^2)}{(1+e(1-t^2)/(1+t^2))^2}\frac{2dt}{1+t^2} =$$

$$= 4\int\limits_0^{+\infty}\frac{(1+e^2)(1+t^2)+2e(1-t^2)}{(1+t^2+e(1-t^2))^2}dt = 4\int\limits_0^{+\infty}\frac{(1+e)^2+t^2(1-e)^2}{(1+e+t^2(1-e))^2}dt =$$

$$= 4\int\limits_0^{+\infty}\frac{(1+e)^2+t^2(1-e)^2}{(1+e+t^2(1-e))^2}dt = 4\int\limits_0^{+\infty}\frac{1+t^2(1-e)^2/(1+e)^2}{(1+t^2(1-e)/(1+e))^2}dt$$

Transition to variable $\tan u = t\sqrt{\frac{1-e}{1+e}} \Rightarrow \frac{du}{\cos^2 u} = dt\sqrt{\frac{1-e}{1+e}}$

$$4\int\limits_0^{+\infty}\frac{1+t^2(1-e)^2/(1+e)^2}{(1+t^2(1-e)/(1+e))^2}dt = 4\sqrt{\frac{1+e}{1-e}}\int\limits_0^{+\pi/2}\frac{1+\tan^2 u(1-e)/(1+e)}{(1+\tan^2 u)^2}\frac{du}{\cos^2 u} =$$

$$= \frac{4}{\sqrt{1-e^2}}\int\limits_0^{+\pi/2}((1+e)\cos^2 u + \sin^2 u(1-e))du = \frac{4}{\sqrt{1-e^2}}\int\limits_0^{+\pi/2}du(1+e\cos 2u) = \frac{2\pi}{\sqrt{1-e^2}}$$

Substituting into the expression for the average:

$$\langle v^2\rangle = \frac{L}{m\tau}\frac{2\pi}{\sqrt{1-e^2}} = \sqrt{GMa}\cdot\sqrt{\frac{GM}{a^3}} = \frac{GM}{a}$$

Method 2:

$$\langle v^2\rangle = \int\limits_0^{\tau}\frac{(GMm)^2}{L^2\tau}(1+e^2+2e\cos\varphi)dt$$

Averaging a constant gives the constant itself, and from the result of the previous part it is clear, that transition to the variable φ gives a factor $1/(1+e\cos\varphi)^2$, collect in the numerator $(1+e\cos\varphi)$:

$$= \langle v^2\rangle = \int\limits_0^{\tau}\frac{(GMm)^2}{L^2\tau}(e^2-1+2(1+e\cos\varphi))dt =$$

$$= \frac{(GMm)^2}{L^2}(e^2-1) + \int_0^{\tau} \frac{(GMm)^2}{L^2\tau}(2(1+e\cos\varphi))dt =$$

$$= \frac{(GMm)^2}{L^2}(e^2-1) + \int_0^{\tau} \frac{(GMm)^2}{L^2\tau}(1+e\cos\varphi)dt =$$

Take the resulting integral using the universal trigonometric substitution:

$$\frac{(GMm)^2}{L^2\tau}\int_0^{\tau}(1+e\cos\varphi)dt = \frac{2L}{m\tau}\int_0^{2\pi}\frac{d\varphi}{1+e\cos\varphi} = \frac{4L}{m\tau}\int_0^{+\infty}\frac{1}{1+e(1-t^2)/(1+t^2)}\frac{2dt}{1+t^2} =$$

$$= \frac{8L}{m\tau}\int_0^{+\infty}\frac{1}{1+t^2+e(1-t^2)}dt = \frac{8L}{m\tau}\int_0^{+\infty}\frac{dt}{1+e+t^2(1-e)} = \frac{8L}{m\tau}\frac{\pi/2}{\sqrt{1-e^2}}$$

Substituting into the expression for the average speed:

$$\langle v^2\rangle = \frac{(GMm)^2}{L^2}(e^2-1) + \frac{4\pi}{m\tau}\frac{L}{\sqrt{1-e^2}} = -\frac{GM}{a} + 2\sqrt{GMa}\cdot\sqrt{\frac{GM}{a^3}} = -\frac{GM}{a} + 2\frac{GM}{a} = \frac{GM}{a}$$

Method 3:

$$\langle v^2\rangle = \frac{2}{m}\left(E + \frac{L}{\tau}\int_0^{2\pi}\frac{d\varphi}{1+e\cos\varphi}\right)$$

Take the resulting integral using the universal trigonometric substitution:

$$\frac{2L}{m\tau}\int_0^{2\pi}\frac{d\varphi}{1+e\cos\varphi} = \frac{4L}{m\tau}\int_0^{+\infty}\frac{1}{1+e(1-t^2)/(1+t^2)}\frac{2dt}{1+t^2} =$$

$$= \frac{8L}{m\tau}\int_0^{+\infty}\frac{1}{1+t^2+e(1-t^2)}dt = \frac{8L}{m\tau}\int_0^{+\infty}\frac{dt}{1+e+t^2(1-e)} = \frac{8L}{m\tau}\frac{\pi/2}{\sqrt{1-e^2}}$$

Substituting into the expression for the average speed:

$$\langle v^2\rangle = \frac{2E}{m} + \frac{4\pi}{m\tau}\frac{L}{\sqrt{1-e^2}} = -\frac{GM}{a} + 2\sqrt{GMa}\cdot\sqrt{\frac{GM}{a^3}} = -\frac{GM}{a} + 2\frac{GM}{a} = \frac{GM}{a}$$

Finally:

$$\langle\dot{E}\rangle = -\frac{\alpha GM}{a}$$

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Answer:

$$\langle\dot{E}\rangle = -\frac{\alpha GM}{a}$$

D9 — 0.40 pts

Using the results of points D1, D3, D8, obtain the dependences of a and e on t, if at time $t = 0$ $a = a_0$, $e = e_0$.

From the result of the previous point:

$$\dot{E} = -\frac{\alpha GM}{a} \Rightarrow \frac{d}{dt}\left(\frac{1}{a}\right) = \frac{2\alpha}{m}\frac{1}{a}$$

$$a = a_0 e^{-2\alpha t/m}$$

Substitute expression for semi-axis in dependence moment momentum from time:

$$L = m\sqrt{GMa(1-e^2)} \Rightarrow \sqrt{a(1-e^2)} = \sqrt{a_0(1-e_0^2)}e^{-\alpha t/m}$$

$$\sqrt{1-e^2} = \sqrt{1-e_0^2} \Rightarrow e = e_0$$

Answer:

$$a = a_0 e^{-2\alpha t/m}$$

$$e = e_0$$

D10 0.10 pts

Find the time of descent of orbit T from height $h_0 = 408$ km to height $h = 400$ km, if parameter $\alpha = 7.17 \cdot 10^{-5}$ kg/s, mass $m = 420$ t, $r_0 = 6378$ km. Express your answer in days.

From the result of the previous point:

$$T = \frac{m}{2\alpha} \ln \frac{r_0 + h_0}{r_0 + h}$$

Answer:

$$T = 40 \text{ days}$$